Business Insights

1. **What is the overall loan default rate?**
   The overall default rate is **93.20%** (932 of 1,000 loans).
2. **Which loan type has the highest default rate?**
   Vehicle loans have the highest default rate by loan purpose at **95.35%** (172 loans).
3. **Which region has the highest default rate?**
   This cannot be determined because the dataset does not contain a Region field.
4. **Does credit score appear associated with loan default?**
   Yes. Defaulted loans have a lower average credit score (**683.26**) than paid loans (**717.57**). This is an association in this sample, not proof of cause.
5. **Does income appear associated with default?**
   Yes. Average annual income is **48,072.04** for defaulted loans and **101,921.63** for paid loans.
6. **Are higher debt-to-income ratios associated with default?**
   In this dataset, the average DTI value is **4.57** for defaulted loans and **0.94** for paid loans. The DTI units are not documented, so the scale should be verified.
7. **Are previous defaults associated with current default?**
   The dataset has no Previous_Defaults field. Delinquencies can be used only as a proxy: the average is **0.71** for defaulted loans and **0.63** for paid loans. This small difference does not establish a strong relationship.
8. **Which customer segment has the highest default rate?**
   Among the available segments, **Loan Grade F** has the highest rate at **98.18%** (55 loans). By loan purpose, Vehicle is highest at **95.35%**.
9. **Which factors appear associated with loan default?**
   In this sample, default is associated with lower average credit scores and income, higher average loan amounts and DTI values, and differences by loan purpose and loan grade.
10. **What pattern can be observed between loan amount and default status?**
    Average loan amount is **217,098.62** for defaulted loans, compared with **134,977.87** for paid loans.
11. **What financial factors should an organization monitor?**
    Monitor credit score, annual income, DTI, loan amount, monthly payment, interest rate, delinquencies, loan purpose, and loan grade. Validate these patterns before using them in lending decisions.
12. **Write 5–10 key business insights.**
    a. The dataset contains 1,000 loans, of which 932 are labeled Default.

b. The observed default rate is 93.20%; this unusually high share should be checked against the source data.
c. Vehicle loans have the highest default rate by purpose (95.35%).
d. Loan Grade F has the highest rate among grades (98.18%), based on 55 loans.
e. Defaulted loans have a lower average credit score than paid loans.
f. Defaulted loans have a lower average annual income than paid loans.
g. Average loan amount is higher among defaulted loans than paid loans.
h. Region and previous-default history cannot be analyzed because those fields are absent.